AI Powered Data Analysis Remote Internship

TEAM - 8

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Dataset Overview

- The given dataset consists of 16 columns (features), and 8558 entries and the features names and types as follows:

Learner Sign Up Date Time	Opportunity Id	Opportunity Name	Opportunity Category	Opportunity End Date	First Name	Date of Birth	Gender
object	**object**	**object**	**object**	**object**	**object**	**object**	**object**

Country	Institution Name	Current/Intended Major	Entry created at	Status Description	Status Code	Apply Date	Opportunity Start Date
object	**object**	**object**	**datetime64[ns]**	**object**	**Int64**	**object**	**object**

- That is changed data type:

Learner Sign Up Date Time	Opportunity Id	Opportunity Name	Opportunity Category	Opportunity End Date	First Name	Date of Birth	Gender
datetime64[ns]	string	string	string	datetime64[ns]	string	datetime64[ns]	string

Country	Institution Name	Current/Intended Major	Entry created at	Status Description	Status Code	Apply Date	Opportunity Start Date
string	string	string	**datetime64[ns]**	string	category	datetime64[ns]	datetime64[ns]

Tools used

Pre-processing Steps

Data Cleaning

- Corrected invalid & noisy values
- Removed redundant columns
- Standardized date & time formats
- Fixed text inconsistencies
- Normalized institution names
- Cleaned personal name fields

Feature Engineering

- Age at Application
- Institution Participation Count

- Corrected invalid & noisy values
- Removed redundant columns

- Age at Application
- Institution Participation Count

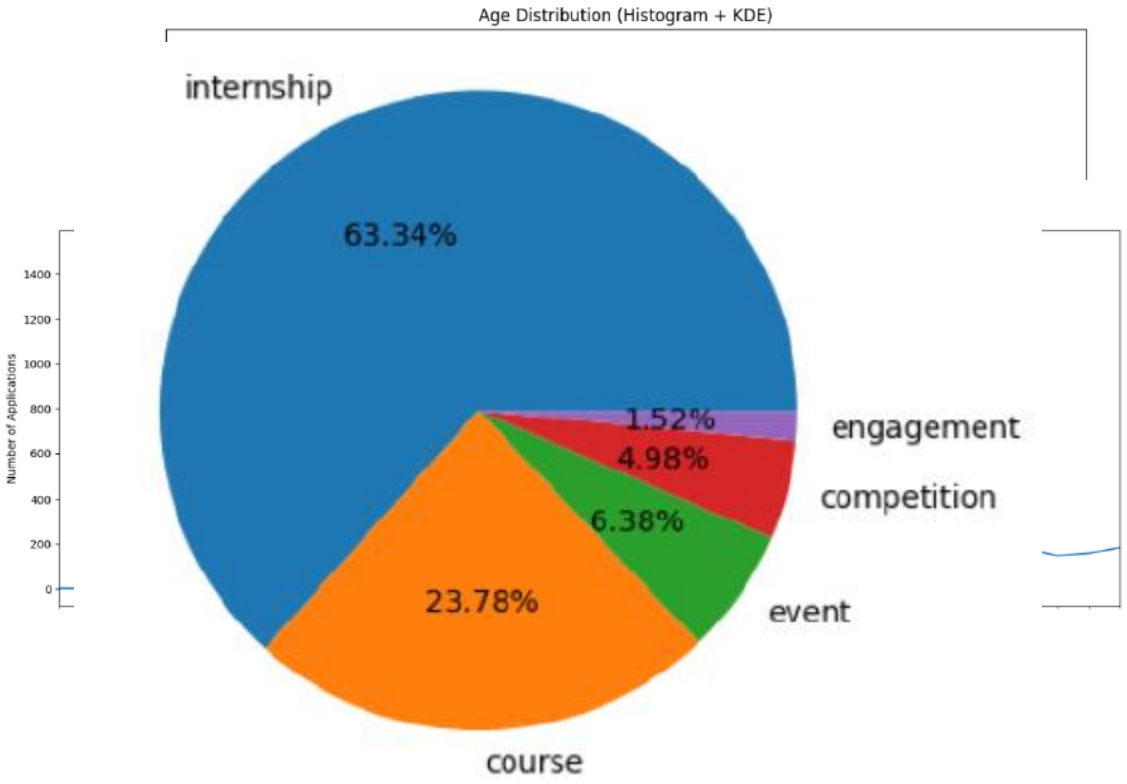
Exploratory Data Analysis

Key Visual Analysis

- Learners by continent and opportunity category
- Gender and age distribution
- Year-wise and month-wise application trends

Major Insights

- North America shows highest participation
- Internships are most popular opportunity
- Majority of learners completed programs
- Strong growth in applications (2023–2024)
- Most active learners aged 18–25



Model Building Link

Feature Selection

Data Split 80/20

Model Training

Evaluation

Trained Models:

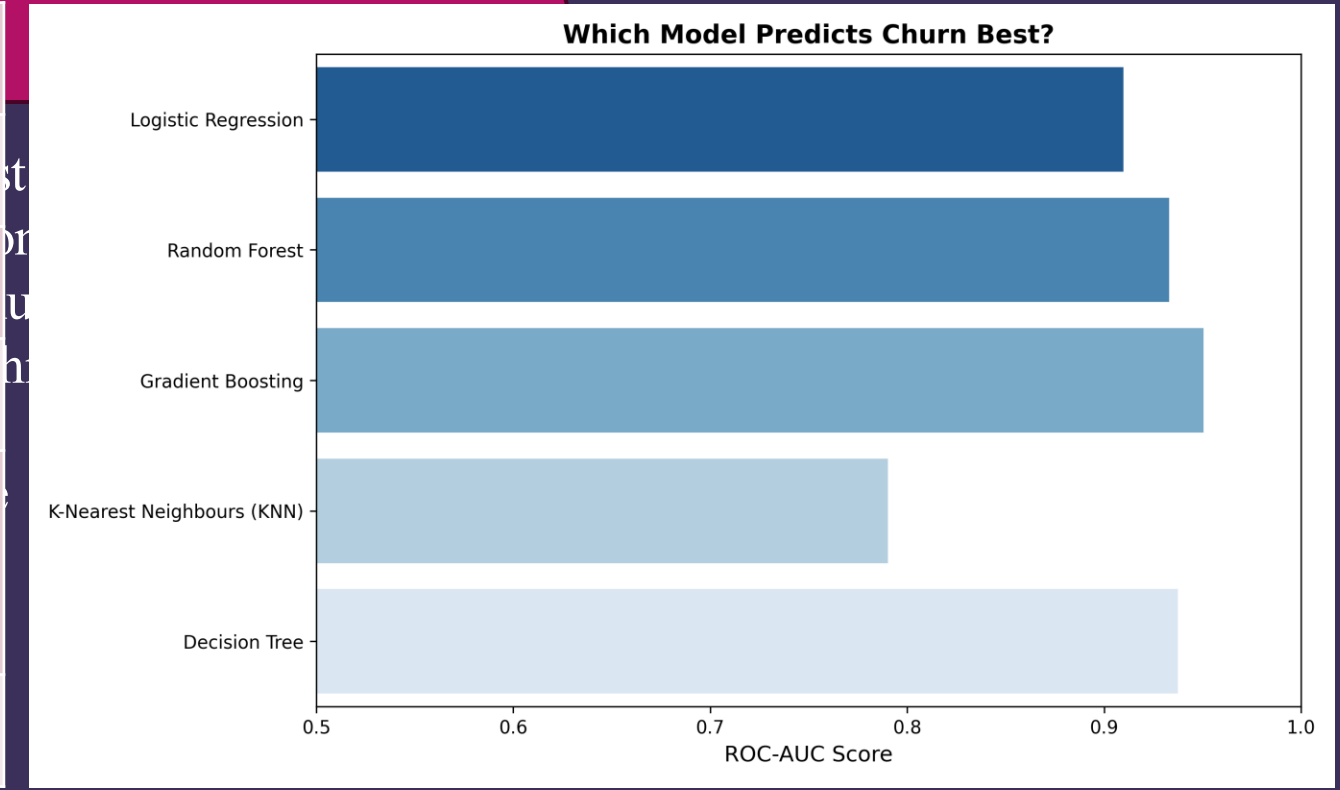
- K-Nearest Neighbors
- Logistic Regression
- Random Forest
- Gradient Boosting
- Decision Tree

Applied stratified train-test split with feature encoding and standardization to ensure robust model performance.

Model Evaluation

Five classification models were trained and evaluated for learner churn prediction. Their performance was compared using accuracy score.

Model	ROC-AUC
Logistic Regression	0.909790257
Random Forest	0.932935806
Gradient Boosting	0.950439716
K-Nearest Neighbours (KNN)	0.790135781
Decision Tree	0.937441527

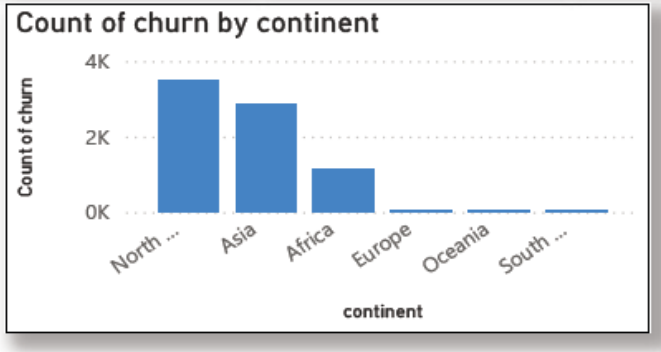


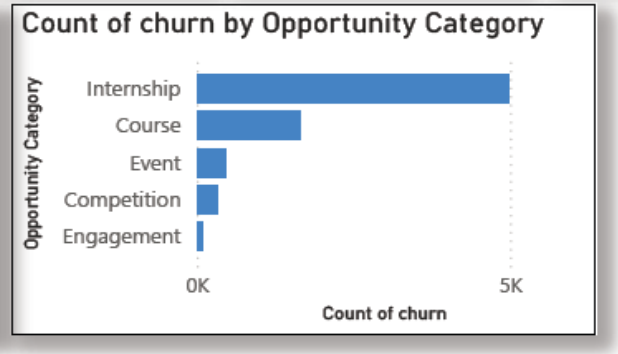
Student Churn Analysis & Engagement Dashboard

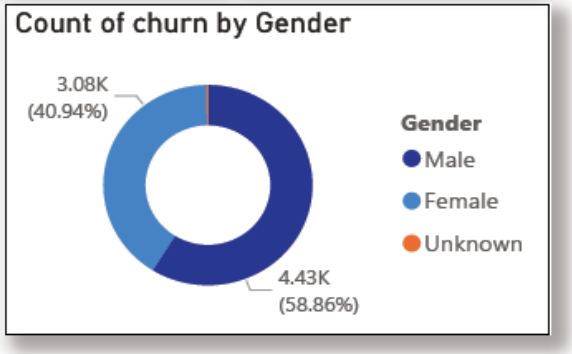
7531	3942	52.3%	90.1%	85.9%	97.0%	91.1%
Total Learners	Total Churned	Churn Rate	Accuracy	Precision	Recall	F1 Score

Opportunity Cat...

- [ ] Competition
- [ ] Course
- [ ] Engagement
- [ ] Event
- [ ] Internship





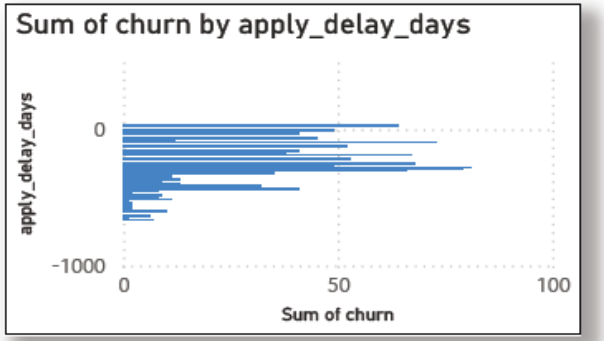


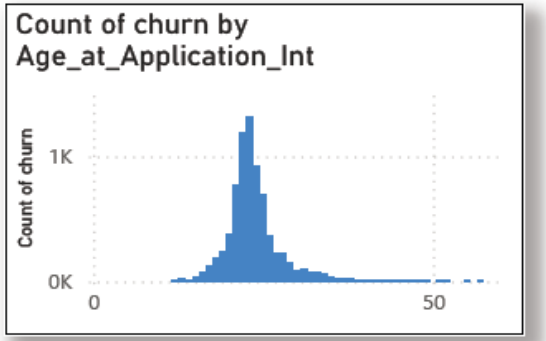
continent

- [ ] Africa
- [ ] Asia
- [ ] Europe
- [ ] North America
- [ ] Oceania
- [ ] South America

Gender

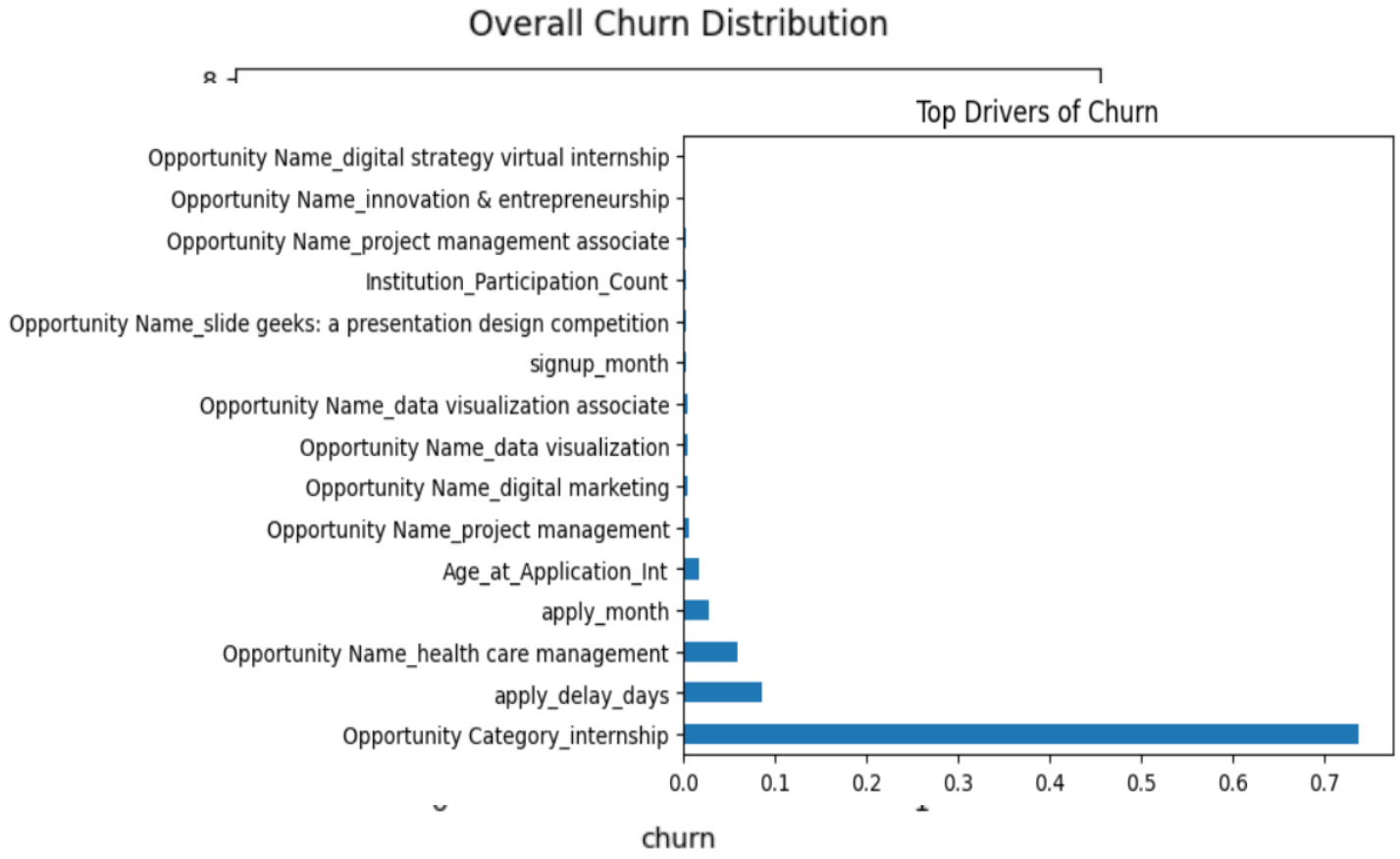
- [ ] Female
- [ ] Male
- [ ] Unknown





Key Factors Affecting Churn

Overall, churn is mainly driven by opportunity type, apply delay days, application , geography, and engagement level rather than demographic variables alone.



Recommendations to Reduce Churn

1. **Enhance Internship Experience**
   Address internship churn rate through better engagement strategies and support system

2. **Strengthen Early Onboarding**
   Reduce days-to-apply metric through faster engagement processes to improve retention

3. **Increase Institutional Support**
   Boost institutional participation levels, which correlate strongly with lower churn rates

4. **Monitor KPIs Continuously**
   Track retention metrics, engagement timing, and completion rates for proactive intervention

Recommendations for reduce churn

Conclusion

- Dataset cleaned, validated and prepared for analysis.
- Key engagement trends identified across regions, demographics, and opportunity types.
- Predictive modeling revealed major churn drivers (age, application timing, participation level).
- Insight supports targeted retention strategies and better learner engagement.
- Project demonstrates practical data analytics workflow from cleaning to prediction.

Internship Reflection

NEW TECHNICAL SKILLS GAINED

Machine Learning Fundamentals

- ✔ Trained & compared multiple classificattion models including Random Forest, Gradient Boosting, Decision Tree, Logistic Regression, and K-Nearest Neighbors for churn prediction

Data Preprocessing & Feature Engineering

- ✔ Cleaned 812 messy records, handled missing values, and created predictive features from raw data

Data Visualization & EDA

- ✔ Uncovered hidden patterns in 8,541 records through exploratory analysis and visual storyteiling

SOFT SKILLS DEVELOPED

Critical Thinking

- ✔ Questioned assumptions, validated outliers, and distinguished correlation from causation, learned how to brainstorm

Communication

- ✔ Translated complex ML findings into clear recommendations for non-technical stakeholders, work as a team and learned to communicate amongst each other

Initiative & Problem-Solving

- ✔ Independently navigated challenges from data cleaning to model selection without hand-holding

Reference Links

Link for the Presentation video

Week 01: Analyse the Data

Week 02: EDA on the Data

Week 03: Model Trainning

thank you!